

Gestational trophoblastic neoplasia is deeply personal to me, and I figured it would honor those in my life who have been affected by it by learning more about the disease. Below, I explored and wrote out what I found:

Introduction

Gestational trophoblastic disease (GTD) encompasses a spectrum of pregnancy-related premalignant and malignant tumor forms that are caused by abnormal proliferation of placental tissue. The malignant forms, known as gestational trophoblastic neoplasia (GTN), include the invasive mole, choriocarcinoma, placental site trophoblastic tumor (PSTT), and epithelioid trophoblastic tumor (ETT) [9]. What makes these tumors unique from other forms of cancer is that they are composed of specialized placenta cells called trophoblasts. In a healthy pregnancy, these cells typically invade the uterine wall to establish a blood supply for the baby [9]. However, in GTN, factors such as abnormal fertilization cause them to aggressively multiply and use the mother's blood vessels to spread to distant organs like the lungs, brain, or liver [6].

How It Begins

The disease often begins when a premalignant condition, most commonly a molar pregnancy, turns malignant [2]. Occurring in 13-20% of complete molar pregnancies, leftover cells from a mole continue to grow after the initial pregnancy has ended, resulting in malignant GTN. However, in partial molar pregnancies, the risk is much lower at only 0.5-5% [6]. Researchers have also identified an abnormal tissue, known as an atypical placental site nodule (APSN), that, although not cancerous itself, can turn into cancerous PSTT or ETT forms [2].

The details behind the growth start with abnormal fertilization. One form of this is a complete mole, which typically occurs when an egg with no active maternal nucleus (empty egg) is fertilized by one sperm that duplicates its DNA after detecting no maternal DNA, or by two sperm, resulting in entirely paternal tissue. A partial mole occurs when a normal egg is fertilized by two sperm, creating a cell with an extra set of chromosomes [9]. Because these tumors contain paternal genetic material, there is essentially foreign tissue within the mother's body.

Epidemiology

GTD occurs in roughly 1 in 1,000 pregnancies globally, though these numbers vary by region. There is a vastly higher incidence in parts of Southeast Asia, where choriocarcinoma occurs in about 23 per 100,000 deliveries, compared to only 3 per 100,000 in the United States and Europe [4][6]. Risk factors for such pregnancies often come from extremes of maternal age and a history of prior molar pregnancies. Additionally, women under 20 or over 35 face double the risk of a sporadic mole, and for those over 40, the risk of developing a molar precursor to GTN goes up to 5.2-7.5 times higher than women in their prime reproductive years [9][6]. Lifestyle factors such as animal fat consumption could potentially be a dietary risk factor that can contribute to the development of the disease, according to research [9].

Tracking the Cancer

GTN is typically diagnosed and monitored through measurements of beta-human chorionic gonadotropin (β -hCG) levels. This pregnancy hormone is secreted by trophoblast tumor cells, acting as a reliable indicator of the magnitude of tumor activity [6]. By tracking these levels weekly, doctors can identify if a

tumor is malignant when the hCG levels plateau or rise. The levels plateau when hCG fails to drop across four measurements over three weeks, and they rise when levels increase significantly across three measurements over two weeks. This strategy allows the cancer to be detected before any masses are visible on a scan or felt by the patient [6].

Common Treatments

GTN is most commonly treated with chemotherapy, but the regimen used is determined by the FIGO 2000 Prognostic Scoring System. It divides treatment into several levels depending on the patient's calculated risk score. Low-risk patients, ranging from 0-6, are treated with single-agent chemotherapy, such as Methotrexate or actinomycin D. It uses a gentle approach with only one type of drug, yielding a cure rate that approaches 100%. As for high-risk patients (score 7-12), a more aggressive multi-drug chemotherapy is used called EMA-CO, combining Etoposide, Methotrexate, Actinomycin-D, Cyclophosphamide, and Vincristine [6].

Side Effects

While these therapies are effective, they come with significant toxicity and side effects. Drugs like Methotrexate can cause nausea and some hair thinning, but EMA-CO that utilizes multiple drugs at once is far more intense. It often leads to alopecia (complete hair loss), severe vomiting, and myelosuppression [8][2]. This means a drop in blood cells, which can leave the patient vulnerable to life-threatening infections, sometimes being the cause of death in some cases. It also causes psychological impacts like reproductive anxiety. Because GTN primarily affects young women, many fear that the drugs utilized may impact their ability to have children in the future [6]. Although standard chemotherapy has generally shown no signs of affecting fertility or the health of future babies, more intense salvage treatments for resistant cases can lead to permanent infertility, adding emotional burden to an already taxing treatment process. Such cases occur when the patient becomes resistant to standard drugs like Methotrexate or EMA-CO, and infertility occurs in nearly 100% of patients who utilize the high-dose chemotherapy (HDC) with peripheral blood stem cell support treatment [2]. Additionally, prolonged exposure to etoposide through combination therapies (like the EMA-CO or EP/EMA regimens) can lead to early menopause, and high doses exceeding 2 grams per square meter (2 g/m^2) are at a much higher risk for permanent infertility [1].

Immunotherapy

As mentioned earlier, trophoblast tumors contain paternal genetic material, meaning that the mother's immune system should naturally attack. However, the tumor survives by overexpressing a protein called PD-L1, which inhibits the mother's T-cells [9]. Because of this discovery, scientists have been utilizing immune checkpoint inhibitors, a group of immunotherapy drugs. The most prominent of these is Pembrolizumab, which blocks the tumor's ability to deactivate T-cells. What makes these drugs so highly praised is their low toxicity on top of their high effectiveness [1]. According to a 2025 meta-analysis, the drug induced complete sustained remission in over 71% of patients who were previously considered untreatable because of multi-drug resistance [1]. Furthermore, immune checkpoint inhibitors have also been seen to be successful in rare subtypes of GTN, like PSTT and ETT, which have historically been difficult to treat [1]. This makes immunotherapy a field for new treatments to be explored in, as its low toxicity can bypass the drawbacks of salvage chemotherapy.

Research Limitations and Gaps

What makes GTN difficult to research is that it only accounts for 1% of all gynecological malignancies [9]. Because of its rarity, much of our current data is based on old records, which can be inconsistent across different research centers where the information is collected [5]. Another gap to be addressed is predicting chemotherapy resistance in low-risk patients. About 20-35% of these women will fail their first single-drug treatment, and researchers aren't fully sure the reason why [3]. Many scientists believe the next breakthrough lies in molecular and genetic biomarkers for deeper insights into understanding how tumors behave[3]. For instance, researchers are studying proteins like HLA-G and PD-L1 to see if their levels can predict resistance to standard chemotherapy. However, they have not been refined enough to be integrated into the official global system for treatment [3].

A report by Lin Jin-Kai et. al noted the critical limitations within the FIGO 2000 system, such as how poorly the scoring system suited the rare subtypes like PSTT and ETT due to being biologically different and less chemosensitive. The article also stated that patients with a higher score within the low-risk category (5 or 6) frequently failed their first single-drug treatment, with resistance rates higher than 60%. Additionally, the researchers suggested the introduction of an “ultra-high-risk” category for cases to address the high mortality rates in patients with massive tumor burdens [3].

FIGO 2026

In July 2026, the International Federation of Gynecology and Obstetrics (FIGO) addressed some of these concerns. FIGO 2026 introduced a separate system for PSTT and ETT, added an intermediate level within the low-risk group for patients with scores of 5 or 6, and added an Ultra-High-Risk tier with a score of 13 or above. For the women who are included in this tier, a gentle induction chemotherapy utilizing low doses of Etoposide and Cisplatin is used. This is due to the phenomenon in which multi-agent chemotherapies, such as EMA-CO, can cause sudden tumor collapse due to their high potency and speed. Tumor collapse often leads to internal bleeding and organ failure, which leads to a high risk of death [7].

Other major changes include the removal of the 6-month rule, which previously required doctors to wait half a year for a diagnosis, but they realized this was dangerous because it allowed aggressive cancers like choriocarcinoma to spread unnecessarily. To standardize imaging measurements, the 2026 rules mandated that only lung metastases measuring 1 centimeter or larger be counted toward a risk score, preventing the possibility of overtreating a tiny spot that may not impact survival [7].

My Connection

In 2011, my mother was diagnosed with Stage IV GTN two years after having children. Because her symptoms were initially overlooked, her beta-hCG levels were in the millions before she requested her own test. What followed was a difficult two-year battle, and she endured almost a year of heavy chemotherapy to beat the disease after showing resistance to the EMA/CO regimen. Her stories of the mental and physical tolls of her battle touched me, so I wanted to understand more about the disease that marked a huge change in her life.

The main reason I wanted to look into GTN was to understand how the landscape for treatment has changed since her diagnosis. Back in 2011, fighting multi-drug-resistant disease meant relying almost entirely on harsh salvage chemotherapies like EMA/EP and undergoing harsh surgeries to remove lung

tumors. But now, modern strategies have been evolving, including a shift toward immunotherapy. Immunotherapy drugs like Pembrolizumab can now treat multiple cases of multi-drug resistance with less toxicity, making the treatment process less arduous.

My mother's resilience and perseverance, along with the expertise of her physicians, made her recovery possible, but ongoing advancements in treatments have paved the way for a future in which patients may have an easier path to recovery.

Citations

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